OSCILLATIONS

Important formulas and concepts directly from the NCERT –

Periodic motion – motion that repeated after certain interval of time.
e.g. uniform circular motion and orbital motion of planets
Oscillatory motion – to and fro motion about a mean position.
" Every oscillatory motion is periodic, but every periodic motion need not to be necessary"
If frequency is small – it is Oscillation
If frequency is high – it is Vibration

- **Period and frequency –**
  - The orbital period of planet mercury is **88 earth days**.
  - The Halley's comet appears after every **76 years**.
  - Frequency of periodic motion,

$$\nu = 1/T$$

unit – s^{-1} or hertz

1hertz = 1Hz = 1oscillation per sec = s-1

- **Displacement** –
  - Function for displacement,

$$f(t) = A \cos \omega t$$

  - Time period,

$$T = 2\Pi / \omega$$

  - Non periodic functions
    - $e^{-\omega t}$
    - $\log(\omega t)$
- **Simple Harmonic Motion** –
  - It is a periodic function in which displacement is a sinusoidal function of time.
  - Displacement x varies with time as,

$$x(t) = A \cos (\omega t + \Phi)$$

  - Angular frequency of SHM,

$$\omega = 2\Pi / T$$

    unit – radian/sec
  - Frequency of oscillations,

$$f = 1/T$$

- **SHM and uniform circular motion (UCM)** –

  - Projection of UCM on diameter of circle follow SHM.

- **Velocity and Acceleration in SHM** –

  - Speed of particle in UCM,

$$\upsilon = \omega A$$

- Instantaneous velocity of a particle executing SHM,

$$\upsilon(t) = -\omega A \sin(\omega t + \Phi)$$

- Negative sign shows that υ has a direction opposite to the positive direction of x-axis.
- Magnitude of centripetal acceleration in UCM,

$$a = \upsilon^2/A = \omega^2 A$$

- Instantaneous acceleration of a particle executing SHM,

$$a(t) = -\omega^2 A \cos(\omega t + \Phi)$$
$$a(t) = -\omega^2 x$$

- Acceleration of particle in SHM is proportional to displacement.
- Direction of acceleration is opposite of displacement
  i.e. for $x>0$, $a<0$ and for $x<o$, $a>0$

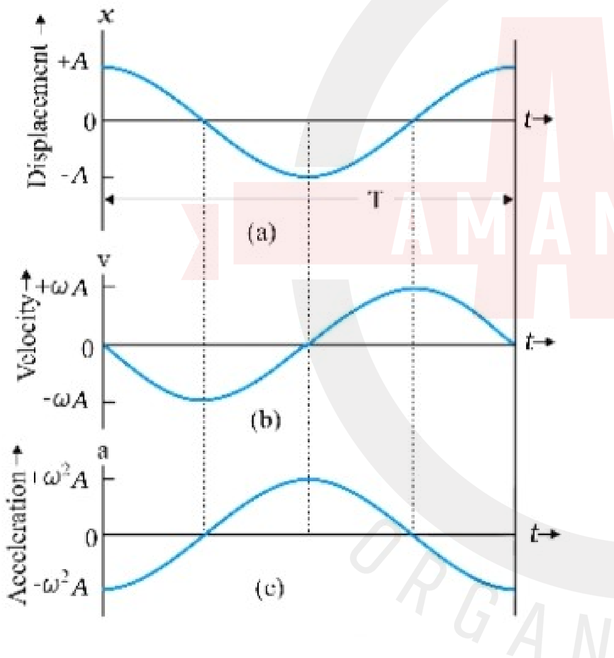

Fig. 13.13 Displacement, velocity and acceleration of a particle in simple harmonic motion have the same period T, but they differ in phase

- With respect to displacement, **velocity** has a phase difference of **Π/2** and **acceleration** has phase diff. of **Π.**

❖ **Force law for SHM** –

- Acc. To newton's second law,

$$\mathbf{F = ma}$$
$$\mathbf{F = -m\omega^2 x}$$
$$\mathbf{F = -kx}$$
$$\mathbf{(k = m\omega^2)}$$
$$\mathbf{\omega = (k/m)^{1/2}}$$

- **Force** is proportional to displacement and directed towards the centre.

Time Period of oscillations –

$$\mathbf{T = 2\Pi\,(m/k)^{1/2}}$$

❖ Energy in SHM –

- Velocity of a particle executing SHM is **zero** at **extreme** positions.
  Similarly kinetic energy of such a particle,

$$K = \tfrac{1}{2} m\upsilon^2$$
$$K = \tfrac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \Phi)$$
$$K = \tfrac{1}{2} kA^2 \sin^2(\omega t + \Phi)$$

  The period of K is T/2.

- Potential Energy possible only for **conservative forces**,
  Spring Force is a conservative force with associated potential energy,

$$U = \tfrac{1}{2} kx^2$$
$$U = \tfrac{1}{2} kA^2 \cos^2(\omega t + \Phi)$$

  It is also periodic with period T/2,being zero at the mean position .

- Total energy,

$$E = K + U$$
$$E = \tfrac{1}{2} kA^2$$

  Total mechanical energy of SHM is independent of time (constant) under any conservative force.

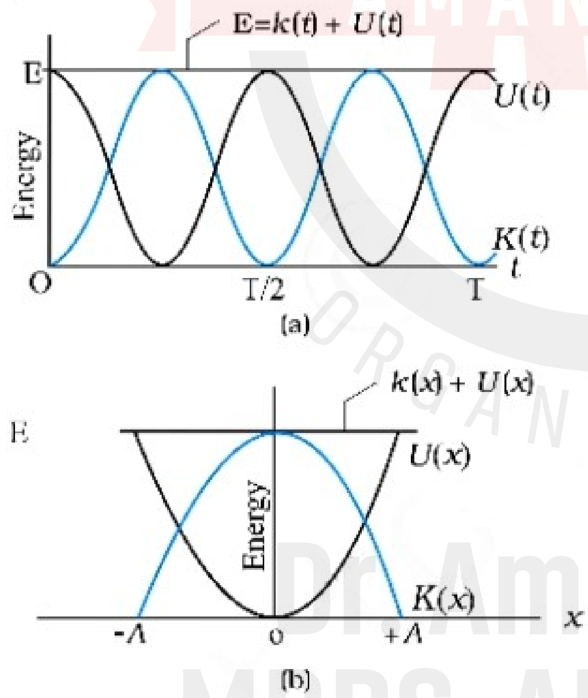

Fig. 13.16 *Kinetic energy, potential energy and total energy as a function of time [shown in (a)] and displacement [shown in (b)] of a particle in SHM. The kinetic energy and potential energy both repeat after a period T/2. The total energy remains constant at all t or x.*

❖ Simple Pendulum –

- **Time period of simple pendulum,**

$$T = 2\Pi(L/g)^{1/2}$$

SUMMARY

1. The motion that repeats itself is called *periodlic motion*.
2. The *period* T is the time reequired for one complete oscillation, or cycle. It is related to the frequency v by,

$$T = \frac{1}{\upsilon}$$

The *frequency* ν of periodic or oscillatory motion is the number of oscillations per unit time. In the SI, it is measured in hertz :

$$1 \text{ hertz} = 1 \text{ Hz} = 1 \text{ oscillation per second} = 1 s^{-1}$$

3. In *simple harmonic motion* (SHM), the displacement $x\ (t)$ of a particle from its equilibrium position is given by,

$$x\ (t) = A \cos\ (\omega t + \phi) \qquad \text{(displacement)},$$

in which A is the *amplitude* of the displacement, the quantity $(\omega t + \phi)$ is the phase of the motion, and ϕ is the *phase constant*. The *angular frequency* ω is related to the period and frequency of the motion by,

$$\omega = \frac{2\pi}{T} = 2\pi\nu \qquad \text{(angular frequency)}.$$

4. Simple harmonic motion can also be viewed as the projection of uniform circular motion on the diameter of the circle in which the latter motion occurs.
5. The particle velocity and acceleration during SHM as functions of time are given by,

$$v\ (t) = -\omega A \sin\ (\omega t + \phi) \qquad \text{(velocity)},$$

$$a\ (t) = -\omega^2 A \cos\ (\omega t + \phi)$$

$$= -\omega^2 x\ (t) \qquad \text{(acceleration)},$$

Thus we see that both velocity and acceleration of a body executing simple harmonic motion are periodic functions, having the velocity *amplitude* $v_m = \omega A$ and *acceleration amplitude* $a_m = \omega^2 A$, respectively.

6. The force acting in a simple harmonic motion is proportional to the displacement and is always directed towards the centre of motion.
7. A particle executing simple harmonic motion has, at any time, kinetic energy $K = ½\ mv^2$ and potential energy $U = ½\ kx^2$. If no friction is present the mechanical energy of the system, $E = K + U$ always remains constant even though K and U change with time.
8. A particle of mass m oscillating under the influence of Hooke's law restoring force given by $F = -k\,x$ exhibits simple harmonic motion with

$$\omega = \sqrt{\frac{k}{m}} \qquad \text{(angular frequency)}$$

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \text{(period)}$$

Such a system is also called a linear oscillator.

9. The motion of a simple pendulum swinging through small angles is approximately simple harmonic. The period of oscillation is given by,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

POINTS TO PONDER

1. The period T is the *least time* after which motion repeats itself. Thus, motion repeats itself after nT where n is an integer.
2. Every periodic motion is not simple harmonic motion. Only that periodic motion governed by the force law $F = -k\,x$ is simple harmonic.
3. Circular motion can arise due to an inverse-square law force (as in planetary motion) as well as due to simple harmonic force in two dimensions equal to: $-m\omega^2 r$. In the latter case, the phases of motion, in two perpendicular directions (x and y) must differ by $\pi/2$. Thus, for example, a particle subject to a force $-m\omega^2 r$ with initial position (0, A) and velocity (ωA, 0) will move uniformly in a circle of radius A.
4. For linear simple harmonic motion with a given ω, two initial conditions are necessary and sufficient to determine the motion completely. The initial conditions may be (i) initial position and initial velocity or (ii) amplitude and phase or (iii) energy and phase.
5. From point 4 above, given amplitude or energy, phase of motion is determined by the initial position or initial velocity.
6. A combination of two simple harmonic motions with arbitrary amplitudes and phases is not necessarily periodic. It is periodic only if frequency of one motion is an integral multiple of the other's frequency. However, a periodic motion can always be expressed as a sum of infinite number of harmonic motions with appropriate amplitudes.
7. The period of SHM does not depend on amplitude or energy or the phase constant. Contrast this with the periods of planetary orbits under gravitation (Kepler's third law).
8. The motion of a simple pendulum is simple harmonic for small angular displacement.
9. For motion of a particle to be simple harmonic, its displacement x must be expressible in either of the following forms :

$$x = A\cos\omega t + B\sin\omega t$$

$$x = A\cos(\omega t + \alpha),\ x = B\sin(\omega t + \beta)$$

The three forms are completely equivalent (any one can be expressed in terms of any other two forms).

Thus, damped simple harmonic motion is not strictly simple harmonic. It is approximately so only for time intervals much less than $2m/b$ where b is the damping constant.

Importants from NCERT summary and points to ponder –

- **The periodic motion governed by the force law, "F = -kx" is SHM.**
- **A combination of two SHMs is only periodic if <u>frequency of one motion is an integral multiple of the other's frequency</u>.**
- **The period of SHM <u>doesn't depend on amplitude or energy or the phase constant.</u>**
- **The motion of simple pendulum is simple harmonic for "<u>small angular displacement</u>"**